Angled launches

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use components to analyze an angled projectile.

SUCCESS CRITERIA - I can...

1. Separate the initial velocity into two components.
2. Use vertical motion to determine flight time.
3. Explain the assumptions behind range comparisons.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “component” and “projectile” to explain your model or result. Sentence starters are optional.

component: the part of a vector along one chosen axis

projectile: an object moving after launch under the stated forces

initial: at the beginning of the interval; subscript i

01 Warm-up: bring back an earlier idea

A downward-velocity fit goes from 0.3 m/s at 0.01 s to 3.24 m/s at 0.31 s.
Find g and explain why fitting an interval can be better than one noisy point.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $v_{x,\,i} = v_i \cos\theta$; $v_{y,\,i} = v_i \sin\theta$; $t_{flight} = 2\,v_{y,\,i}/g$ range = $v_{x,\,i}\,t_{flight}$.
θ above horizontal; g = 10 m/s². Equal launch/landing height; no air resistance.

Worked reasoning: A projectile has initial components 4 m/s horizontal and 10 m/s upward. With g=10 m/s² it reaches the top in 1 s and returns to launch height in 2 s; range=8 m.

Your turn: Sketch its path and velocity arrows at launch, top and return.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

At the highest point, which velocity component is zero and which remains? Is acceleration zero?

Sentence starter: I notice … . This supports … because … .



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3

04 Apply the physics

A ball launches from level ground at 20 m/s, 30° above horizontal, and lands at the same height.
Use g = 10 m/s² and ignore air resistance.
Draw the initial velocity components.
Find flight time, range and maximum height.

Sentence starter: I found … [unit]. This value means … in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3

05 Explain your reasoning

Why is doubling time to the top valid only when the projectile returns to launch height in this ideal model?

Word help: ideal = using stated simplifying conditions

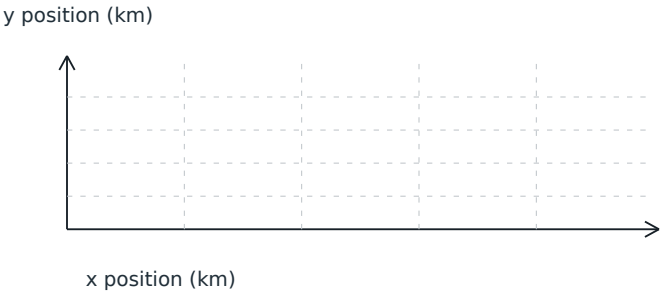
Sentence starter: Under the assumption …, the evidence … supports … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A simplified model predicts position (-70, 90) km with bounds ±7 km in x and ±9 km in y. Write a short briefing giving both intervals and two reasons it is not a real asteroid collision forecast.
Label the given and predicted quantities on the model.

Sentence starter: I predict … . My model supports this because … .



Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

For speed 20 m/s, compare ranges at 30° and 60° on level ground.
Explain which has longer flight time.
Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.

Sentence starter: Both … . They differ in … because … .

Separate horizontal and vertical motion

Label x and y; state positive directions.

Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3

08 STRETCH: test the model

For a launch at 20 m/s from level ground, find the two angles giving an ideal range of 20 m with g=10 m/s².
Compare flight times.
Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.

Word help: ideal = using stated simplifying conditions

Sentence starter: Both … . They differ in … because … .



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3